# Instrumentation, Data analysis and Machine Learning

## Master in Astrophysics, Particle Physics and Cosmology

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# Content

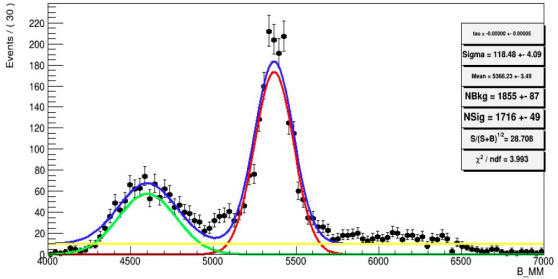
Instrumentation

Data analysis

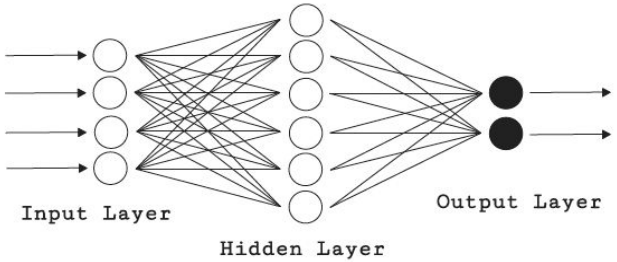
Machine learning



Joan Oro telescope (CAM)




Credit: A. Santander




Credit: V. Costa

# General information

- Optional subject (2nd semester), joint astro & particle & cosmology
  - 6 credits = 150h
  - Lectures: 12h (first weeks)
  - Practical sessions: 22h
  - Workshops and field trips: 18h
  - Supervised project: 50h
  - Independent learning: 48h

- Professors
  - Astrophysics: P. Bordas and X. Luri
  - Gravitational waves: TBC
  - Particle physics: C. Marin

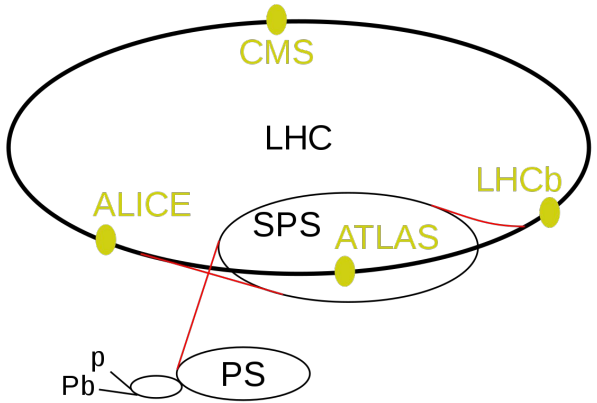
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Dpt. Física Quàntica i Astrofísica

# Lectures

First weeks (12h):

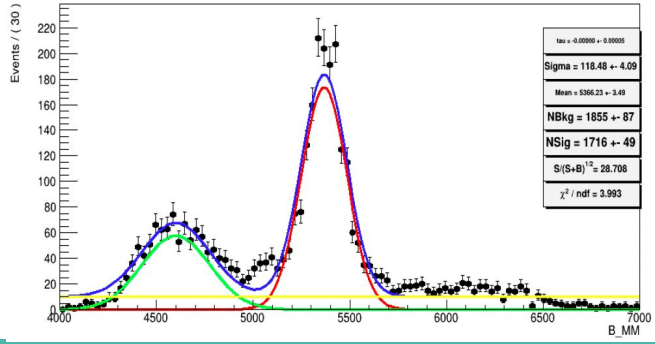
1. Requirements of particle physics experiments
2. Requirements of astrophysical observations
3. Particle accelerators
4. Detection techniques
5. Design of high energy physics experiments
6. Data acquisition and processing
7. Astrophysical instrumentation
8. Astrophysical observation techniques

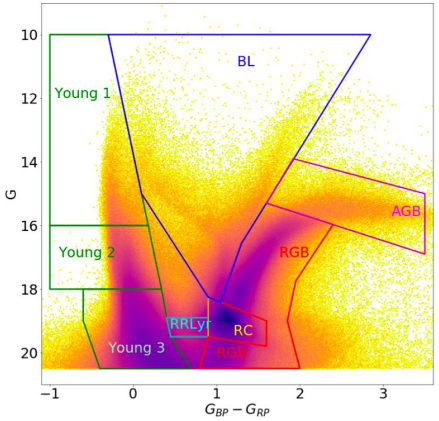


# Practical sessions

During all the course (22h):

- Machine learning and fitting techniques using LHCb data
- Cloud computing using Gaia data
- X-rays using Chandra and XMM-Newton data
- Gamma rays using Fermi-LAT and CTA data
- Gravitational waves using Virgo data



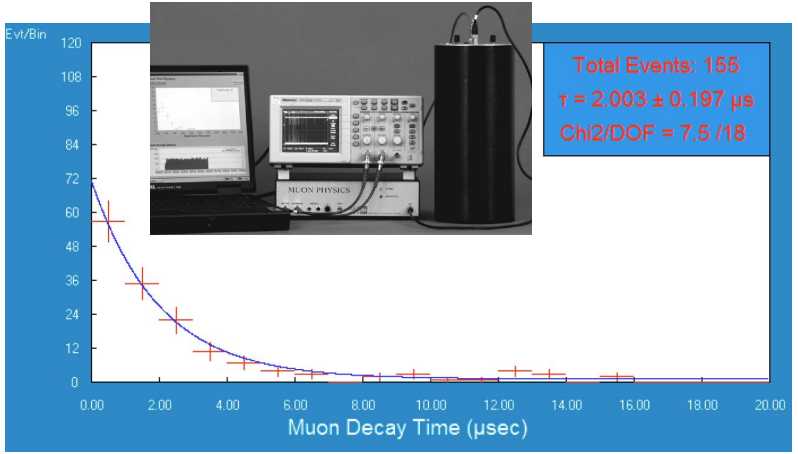
# Workshops and field trips (1)

During all the course (18h):

## ALBA synchrotron beam emittance



## Muon lifetime




Credit: J. R. Stocchetti

# Workshops and field trips (2)

OPTIONAL

During all the course (18h):

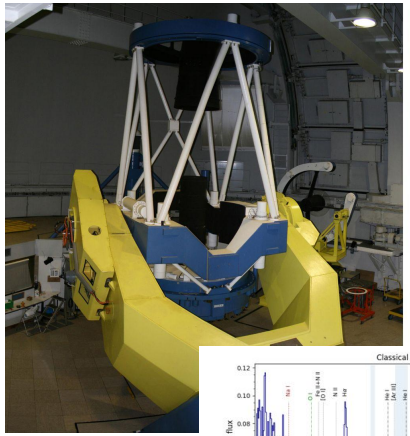
Observation at

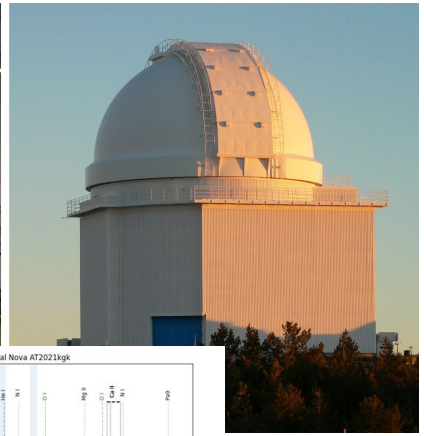
**Parc Astronòmic del Montsec**

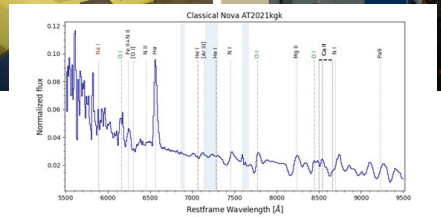


Proposal and observation at

**Calar Alto Astronomical Observatory**



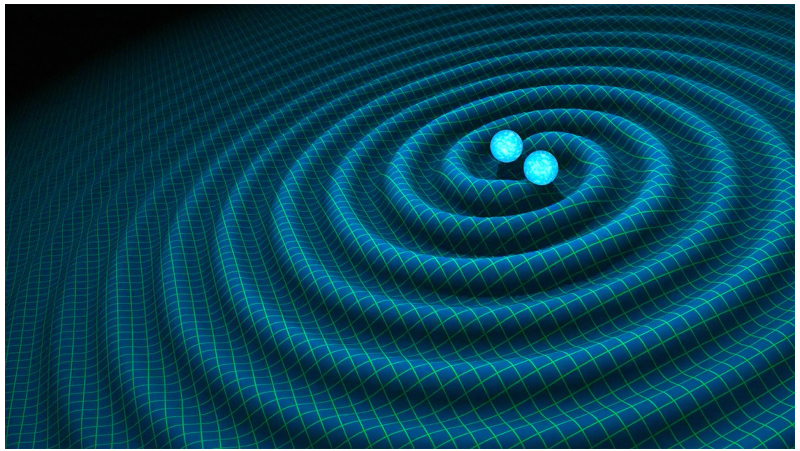





*Extinction-corrected spectrum of the galactic nova AT2021kgk obtained with instrument CAFOS attached to the 2.2 m Zeiss reflector of Calar Alto, in observing time of the Calar Alto Academy program assigned to practical work for the University of Barcelona. Credits: Nadia Blagorodovna.*

# New this year: Gravitational waves

- Topic covered in astrophysical instrumentation lecture
- Dedicated hands-on session using Virgo data by T. Andrade
- Topics will also be proposed for final supervised project



# Supervised project (50h)

To be submitted at the end of the course (but 50h of work, so start early!)

Related to a topic of the subject or to your TFM project (extension)

Topic examples:

- Related to site testing, instrument characterisation or telescope setting up
- Calibration or time alignment of detectors
- LHCb, Fermi satellite, MAGIC telescope, Gaia or LIGO-Virgo data analysis
- Radio interferometric observations

Examples from last course:

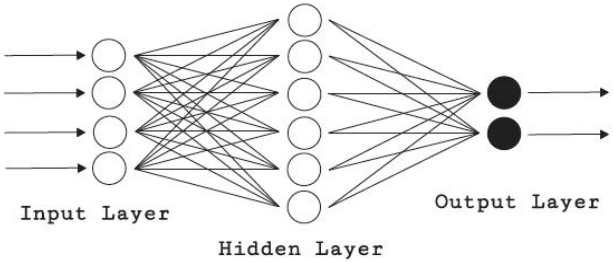
- *Simulation of very eccentric BBH coalescence and gravitational lensing effects*
- *Optimisation and tuning of a BDT*

# Summary

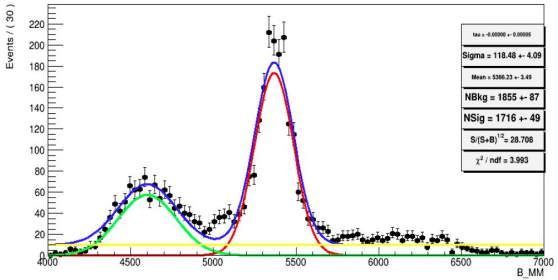
Unique opportunity to access instrumentation and data from ongoing Particle Physics, Astrophysics and Gravitational waves experiments!



Joan Oro telescope (CAM)




Credit: V. Costa




Credit: A. Santander

# Thanks for the attention

